

Magnetni disk in SSD

Primož Lah, 2. letnik Ma-Ra, Pedagoška fakulteta UL

Predmet: Arhitekture računalniških sistemov
Predavatelj: dr. Matej Zapušek

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Magnetni disk in SSD

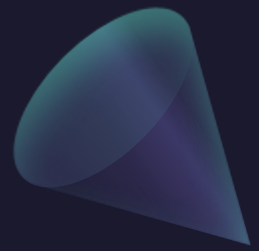
Zgradba in delovanje

Običajne izvedbe

Postopek namestitve

Primerjava specifikacij











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Magnetni disk

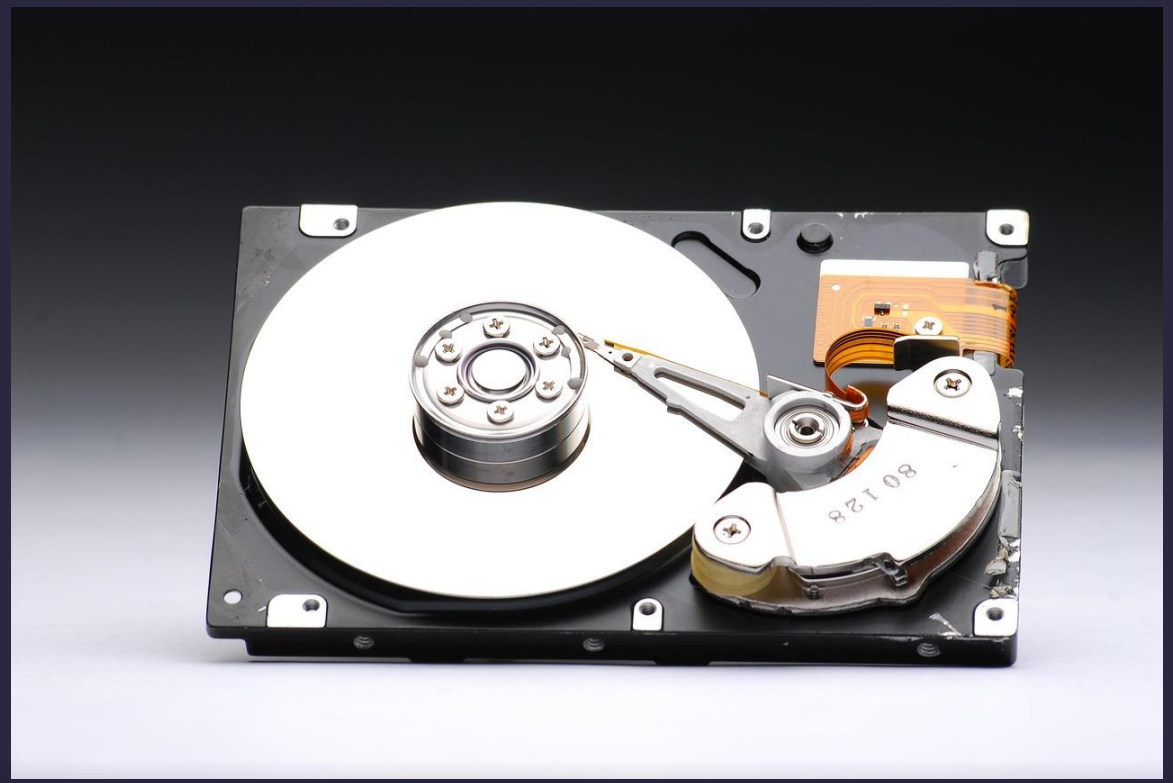
oz. HDD



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Zgradba in delovanje



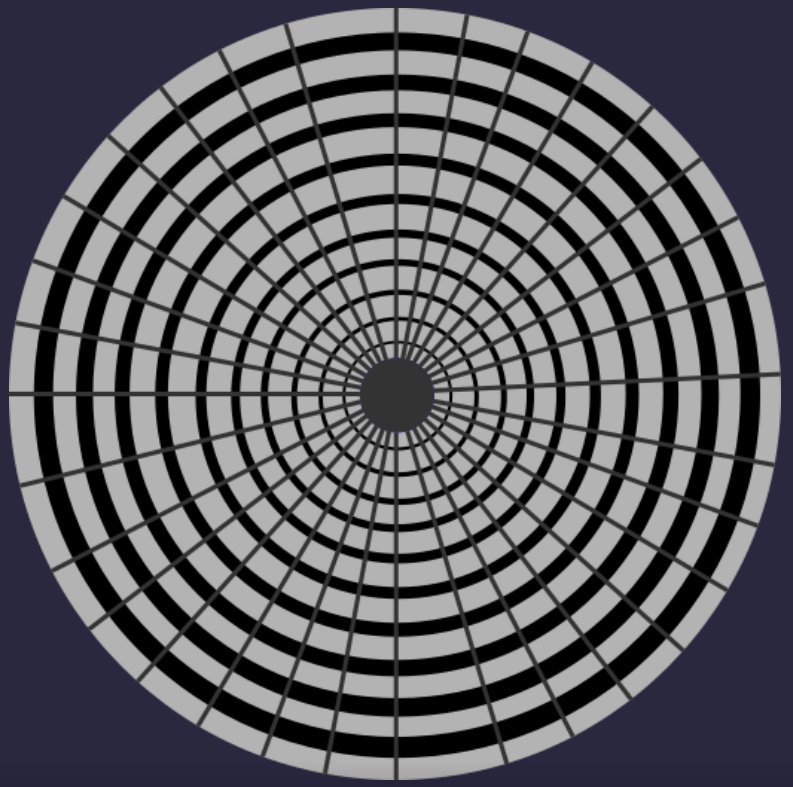
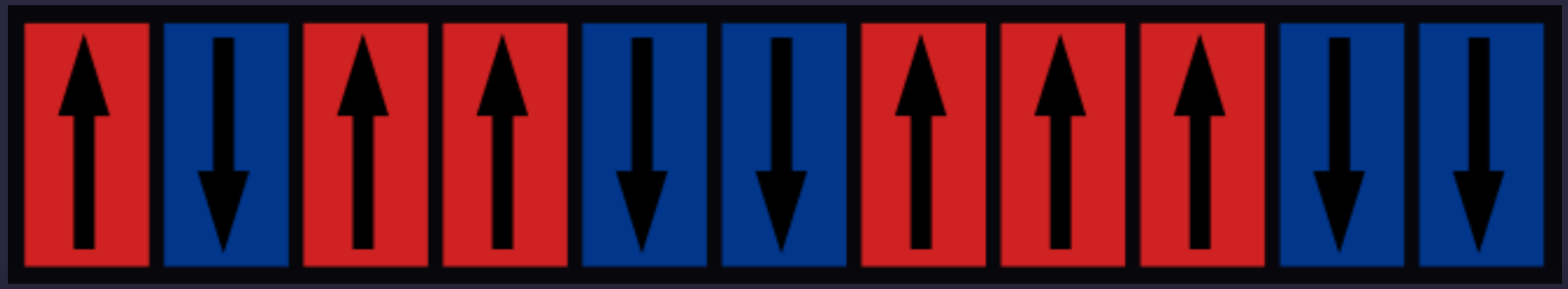


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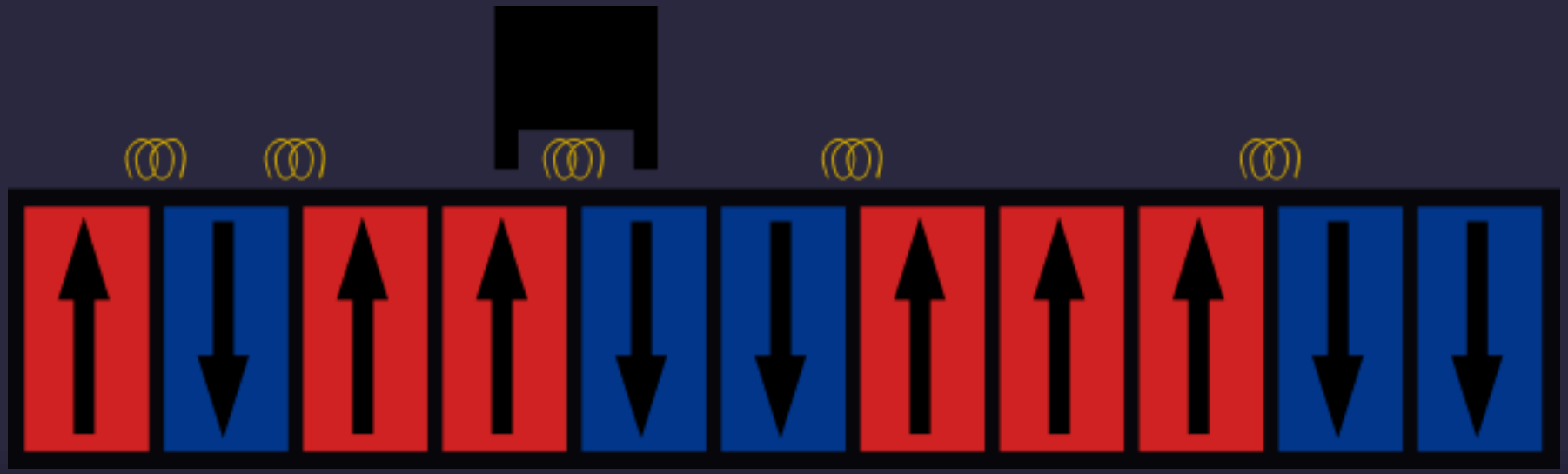
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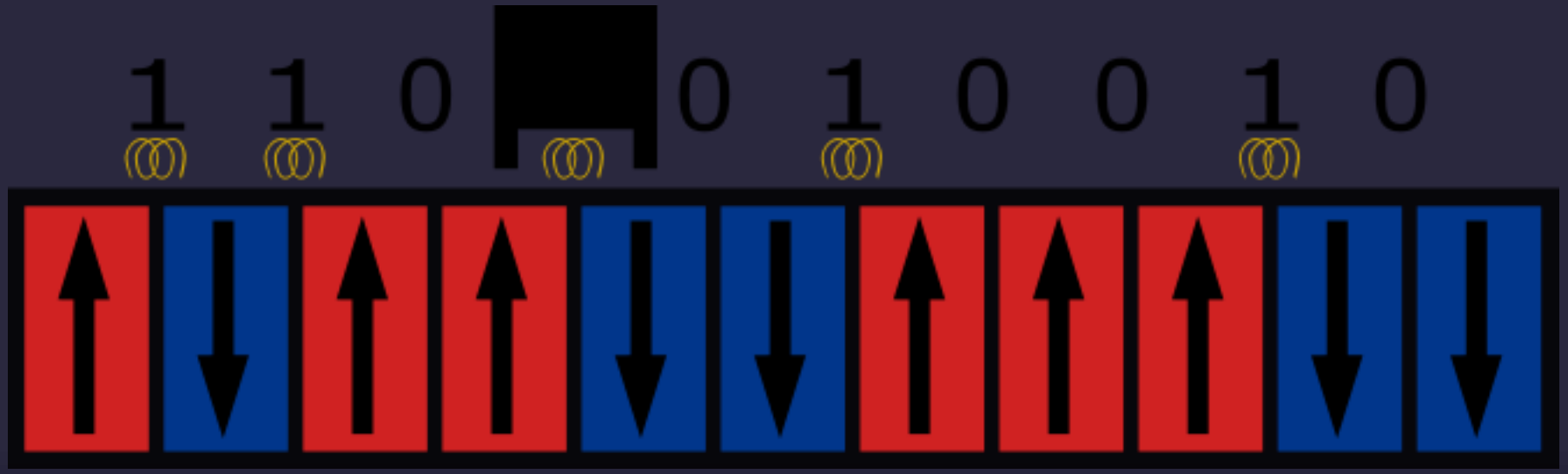
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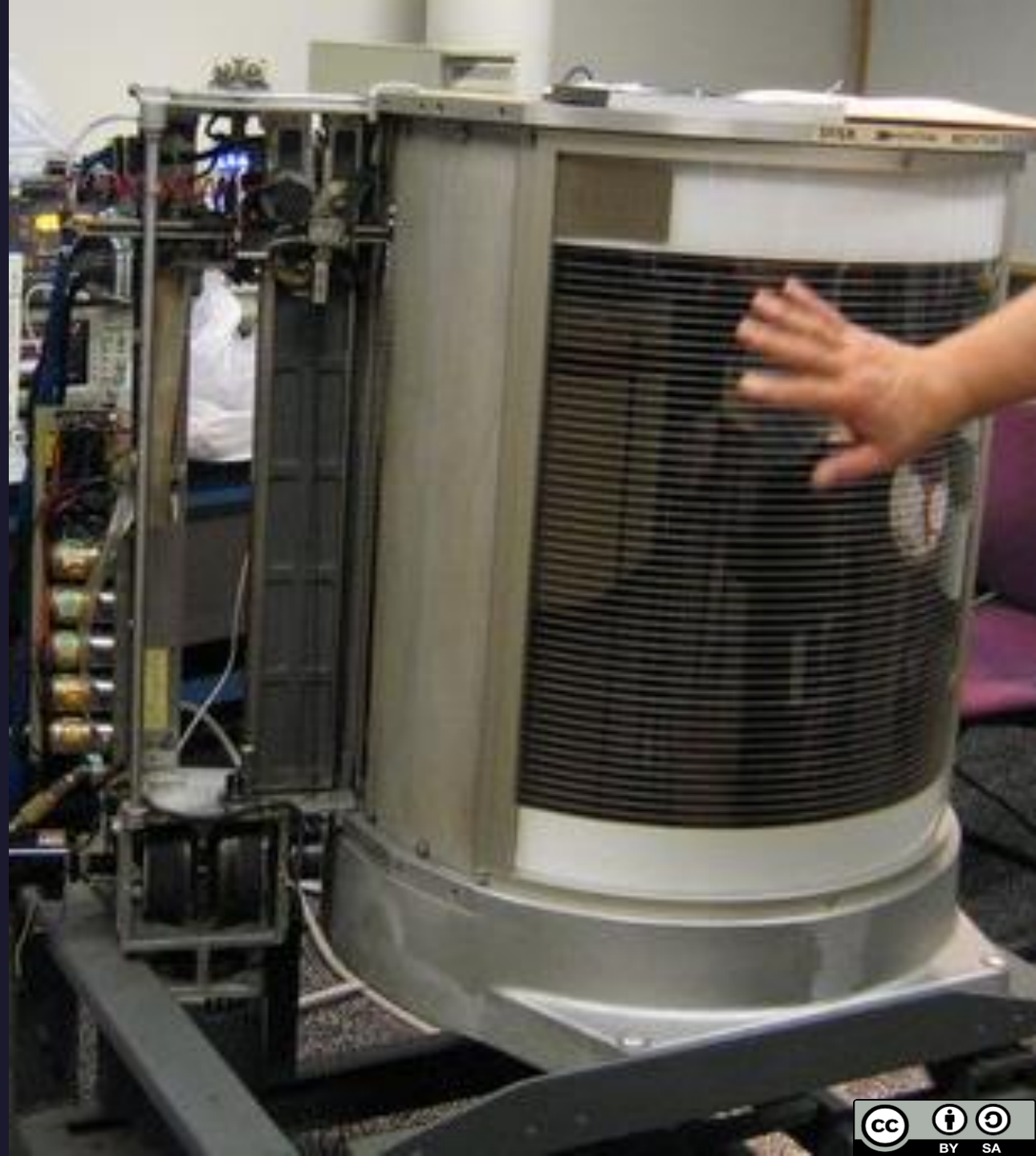
IBM 350

Prvi magnetni disk, leta 1956

52 diskov, 24"

1 bralno-pisalna glava

Skupno 3.75 MB



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Velikosti magnetnih diskov

Naziv velikosti	Dimenzije (d x š x v)	Največja kapaciteta
3.5"	146 x 101 x 20 mm	36 TB (januar 2025)
2.5"	101 x 69 x 10 mm	6 TB (maj 2024)
1.8"	78 x 54 x 5 mm	320 GB (2009)
1.3"	50 x 36 x 10 mm	40 GB (2007)
1"	42 x 36 x 5 mm	12 GB (2006)
0.85"	32 x 24 x 5 mm	8 GB (2004)
5.25" (Half Height)	203 x 146 x 41 mm	19 GB (1998)
5.25" (Full Height)	203 x 146 x 82 mm	47 GB (1998)
8"	368 x 241 x 117 mm	2 GB (1989)

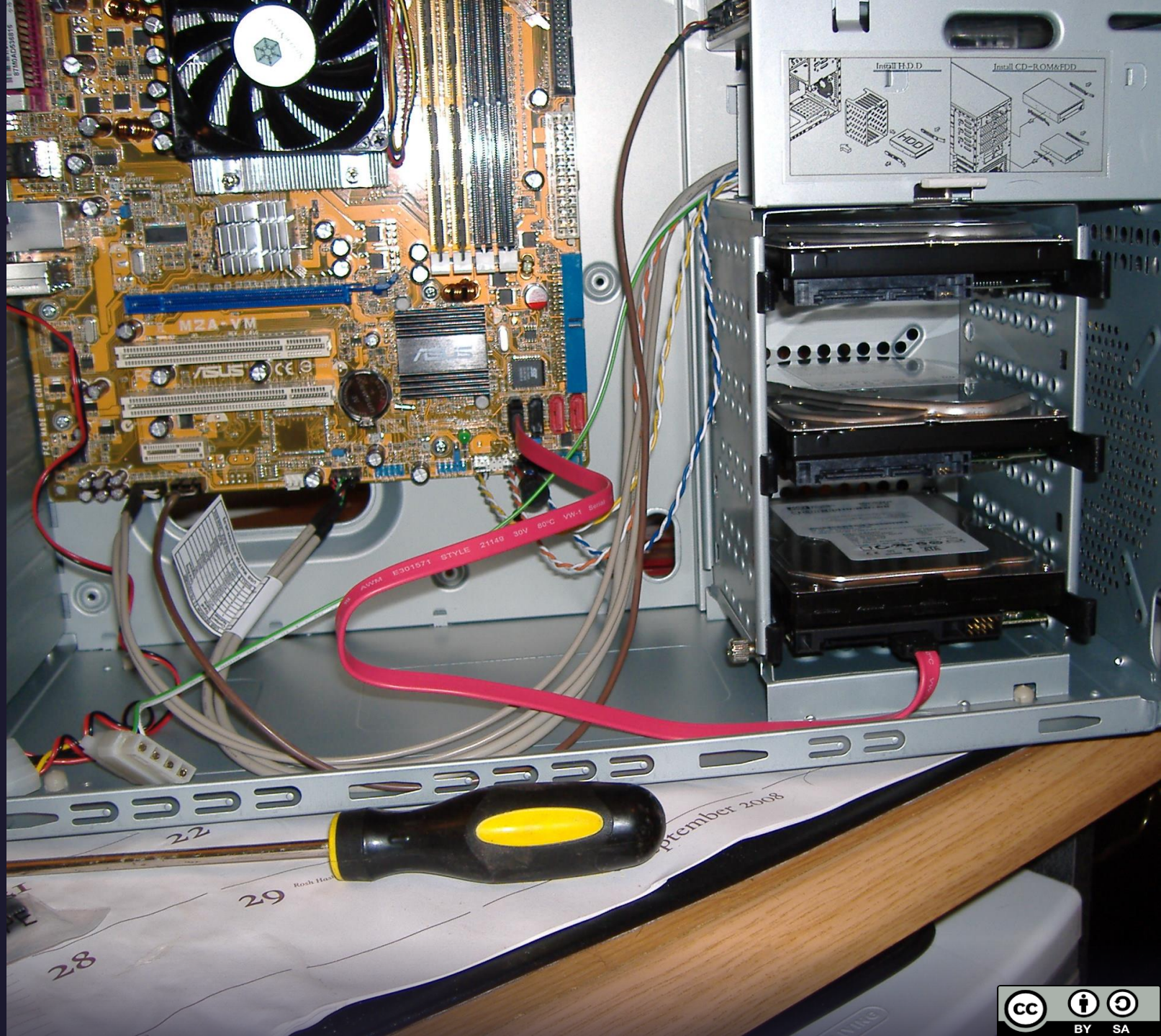
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Namestitev



"New Computer" by Steve W, CC BY-SA 2.0

Negibljivi disk

oz. SSD

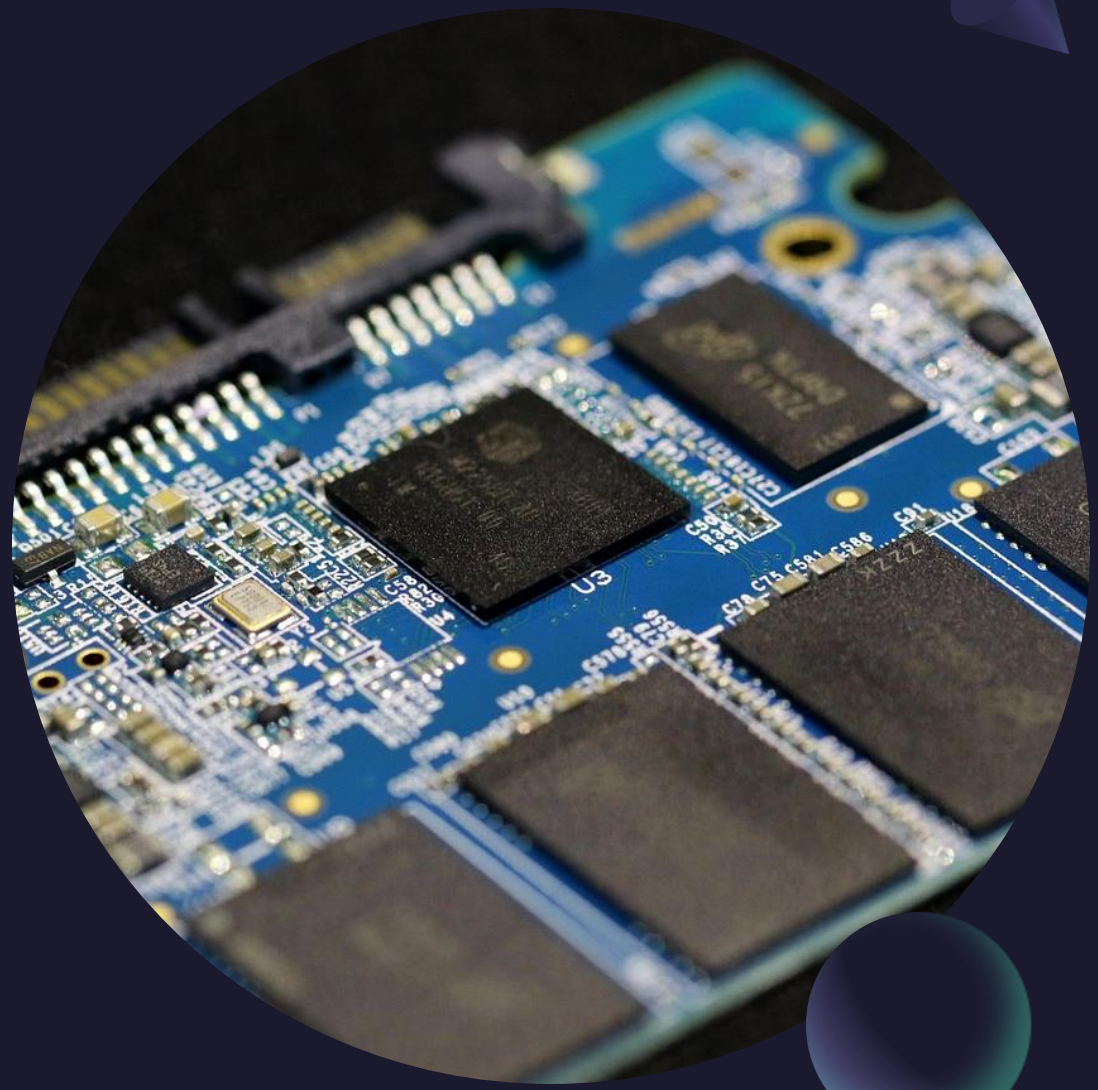


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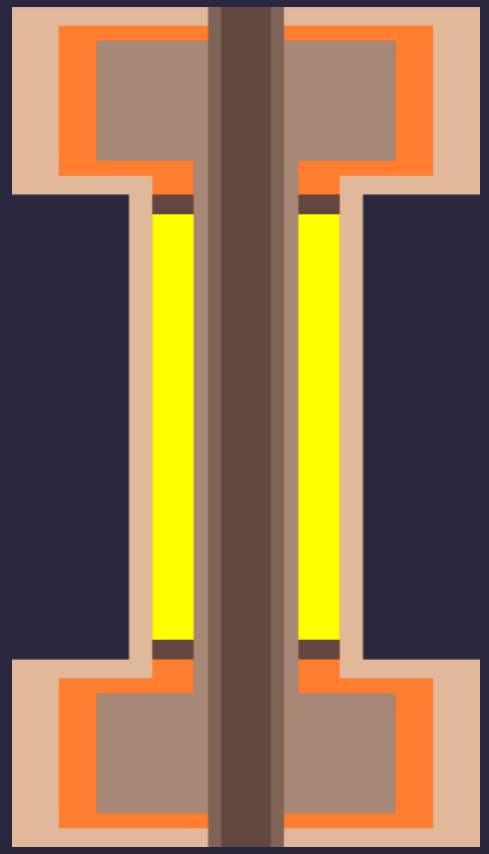


- Nabor podatkovnih celic:
  - Višina: 102 celic
  - Širina: 104 celic
  - Globina: 104 celic
- 8 takšnih enot v enem čipu

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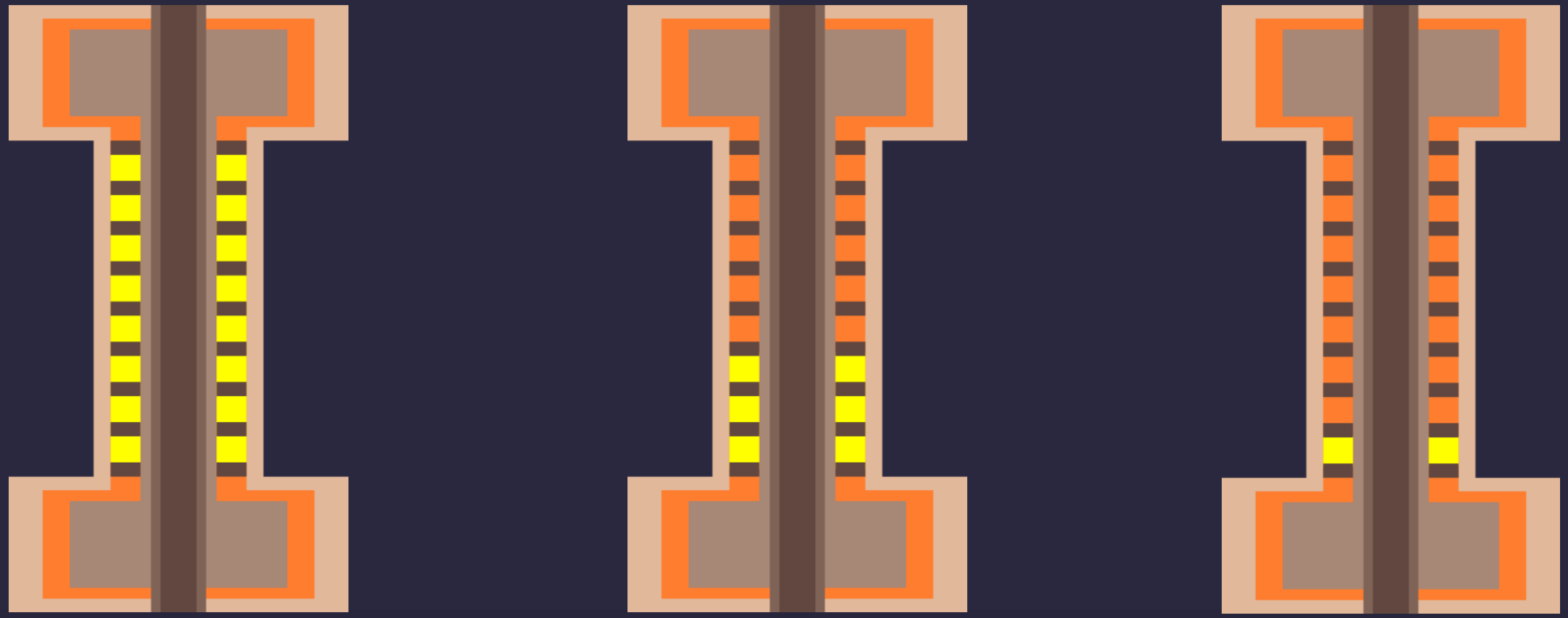
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Oblike SSD

SATA



- PCI-E NVME



"SUPER★TALENT FTM60GK25H Rev.2 SSD 60GB" by Yusuke Kawasaki, CC BY 2.0

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Namestitev



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Primerjava HDD in SSD

Lastnosti	HDD	SATA SSD	PCI-e NVMe SSD
Hitrost	200 MB/s	Do 750 MB/s	Do 4000 MB/s
Kapaciteta	Običajno do 4 TB, lahko do 20 TB Največ 36 TB	Običajno do max. 4 TB Največ 25 TB	Običajno do max. 4 TB Največ 25 TB
Količina predpomnilnika	Od 8 do 256 MB	/	/
Cena za GB	0.05-0.10€/GB (pri 1 TB) 0.025-0.05€/GB (pri 4 TB)	0.10€/GB (pri 1 TB) 0.075€/GB (pri 4 TB)	0.10-0.20€/GB (pri 1 TB) 0.06-0.12€/GB (pri 4 TB)

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Viri

- Branch education: How do Hard Disk Drives Work?
- Branch education: How do SSDs Work?
- TED-Ed: How do hard drives work? - Kanawat Senanan
- Wikipedia: Hard Disk Drive, Solid-state-drive, Magnetic storage, List of disk drive form factors
- Seagate.com
- Mlacom.si
- Anni.si
- Techtarget.com
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